H. J. Thim Trust's

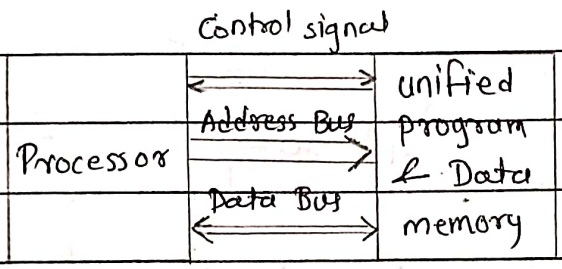
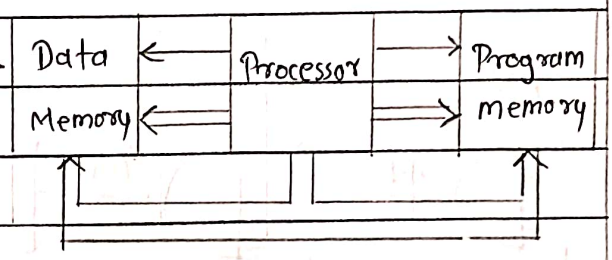
THEEM COLLEGE OF ENGINEERING

Assignment No. 1

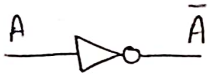
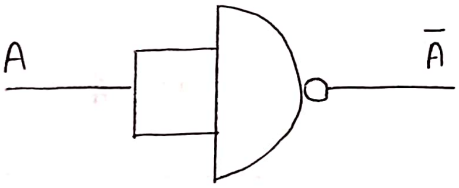
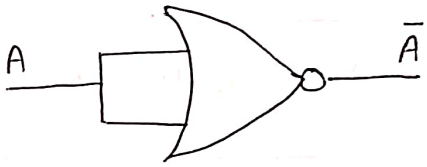
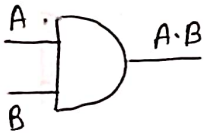
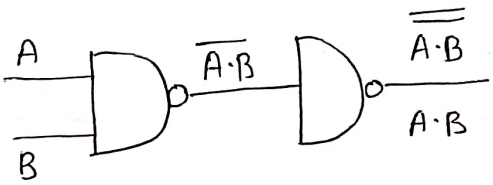
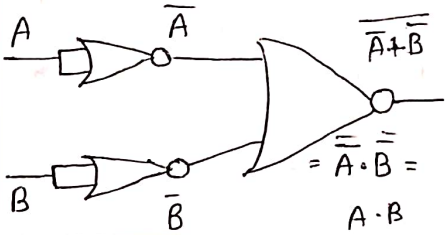
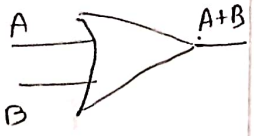
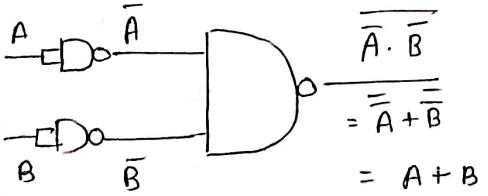
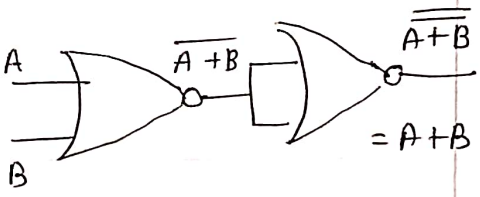
Date :

1] Compare von-neumann and harvard architecture.

→

	Von-neumann	Harvard
		
1)		
2)	In this architecture only one memory is used to store program and data	In this architecture separate memory is used for program and data.
3)	The program and data fetch sequentially. Hence, the execution is slower.	The program and data fetch simultaneously. Hence, execution is faster.
4)	This type of architecture uses CISC machine.	This type of architecture uses RISC machine.
5)	It supports lower degree of parallelism.	It supports higher degree of parallelism.

2] Realise Basic Gates using universal gate only.

Basic Gates	Universal Gates	
	NAND	NOR
NOT 		
AND 		
OR 		

3] i) Convert $(199.375)_{10}$ into binary number.

→ $(199.375)_{10}$

$\frac{199}{2} = 99.5 = 1$

$\frac{99}{2} = 49.5 = 1$

$\frac{49}{2} = 24.5 = 1$

$24/2 = 12 = 0$

$12/2 = 6 = 0$

$6/2 = 3 = 0$

$3/2 = 1.5 = 1$

$1 = 1 = 1$

$0.375 \times 2 = 0.75 = 0$

$0.75 \times 2 = 1.5 = 1$

$0.5 \times 2 = 1 = 1$

$\therefore (199.375)_{10} = (11000111.011)_2$

ii) Convert $(532.125)_8$ into decimal value.

→ $(532.125)_8$

$= 5 \times 8^2 + 3 \times 8^1 + 2 \times 8^0$

$= 346$

$= 1 \times 8^{-1} + 2 \times 8^{-2} + 5 \times 8^{-3}$

$= 0.166015625$

$= 346 + 0.166015625$

$= 346.166015625$

$\therefore (532.125)_8 = (346.166015625)_{10}$

iii) Convert $(110101.101010)_2$ into octal number.

→ $(110101.101010)_2$

$$\underset{6}{\underset{\downarrow}{\underline{110}}}\ \underset{5}{\underset{\downarrow}{\underline{101}}}\ .\ \underset{5}{\underset{\downarrow}{\underline{101}}}\ \underset{2}{\underset{\downarrow}{\underline{010}}}$$

$\therefore (110101.101010)_2 = (65.52)_8$

iv) Convert $(1473.45)_{10}$ into hexadecimal number.

→ $(1473.45)_{10}$

$\frac{1473}{16} = 92.0625 = 0.0625 \times 16 = 1$

$\frac{92}{16} = 5.75 = 0.75 \times 16 = 12$

$5 \longrightarrow 5$

$0.45 \times 16 = 7.2 = 7$
$0.2 \times 16 = 3.2 = 3$
$0.2 \times 16 = 3.2 = 3$
$0.2 \times 16 = 3.2 = 3$

$\therefore (1473.45)_{10} = (5C1.7333)_{16}$

4) Explain derived gate.

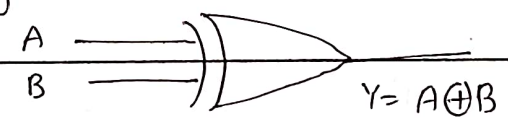
→ - Derived logic gates are the gates that utilize combinations of basic gates (AND, OR, NOT) to perform specific logical operations.

- There are two derived gates such as

i) XOR Gate :-

- An XOR gate produces a high output (1) only when its inputs are different. If both inputs are same (both 0 or both 1), the output is 0.

- Symbol :

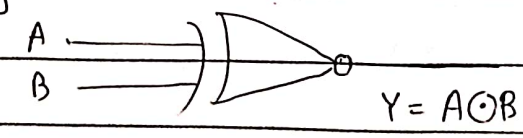


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

ii) XNOR Gate :-

- An XNOR gate produces a high output (1) only when its input are the same (both 0 or both 1). If the inputs are different, then output is 0.

- Symbol :-



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

5] Explain 2's complement in detail.

→ Two's complement :-

- Two's complement is a method of representing signed (positive and negative) binary numbers in computers.
- It allows for straight-forward arithmetic operations (addition and subtraction) using the same circuits for both positive and negative numbers.

- The most significant bit (MSB) indicates the sign (0 for positive and 1 for negative), and the remaining bits represent the magnitude

i) Representing positive numbers.

- Positive numbers are represented in standard binary form.
- for eg, in a 4-bit system, the number 5 would be 0101.

ii) Representing negative numbers :

- To find the two's complement of a negative numbers, you first find the one's complement by inverting all the bits (0's become 1's and 1's become 0's). Then add 1 to the result of one's complement.
- for eg, to represent -5 in a 4-bit system

~~5 → 0110 → 1010~~

5 → 0101

1010 - 1's

1

1011 - 2's

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Assignment No. 2 Date :

1] Multiply $(14)_{10}$ and $(-5)_{10}$ using booth algorithm.

→ $14 \rightarrow (01110)_2 = Q_0$

$-5 \rightarrow (00101)_2 \rightarrow (11010)_2$ - 1's

-70 $\quad 1$

$M = (11011)_2$ - 2's

$\therefore M = -5 = (11011)_2$

& $Q_0 = 14 = (01110)_2$

A	Q_0	Q_{-1}	n	Remark
00000	01110	0	5	Initialize
00000	00111	0	4	ASR
00101	00111	0	3	A=A-M
00010	10011	1		ASR
00001	01001	1	2	ASR
00000	10100	1	1	ASR
11011	10100	1	0	A=A+M
11101	11010	0		ASR

1110111010

0001000101

1

$(0001000110)_2$

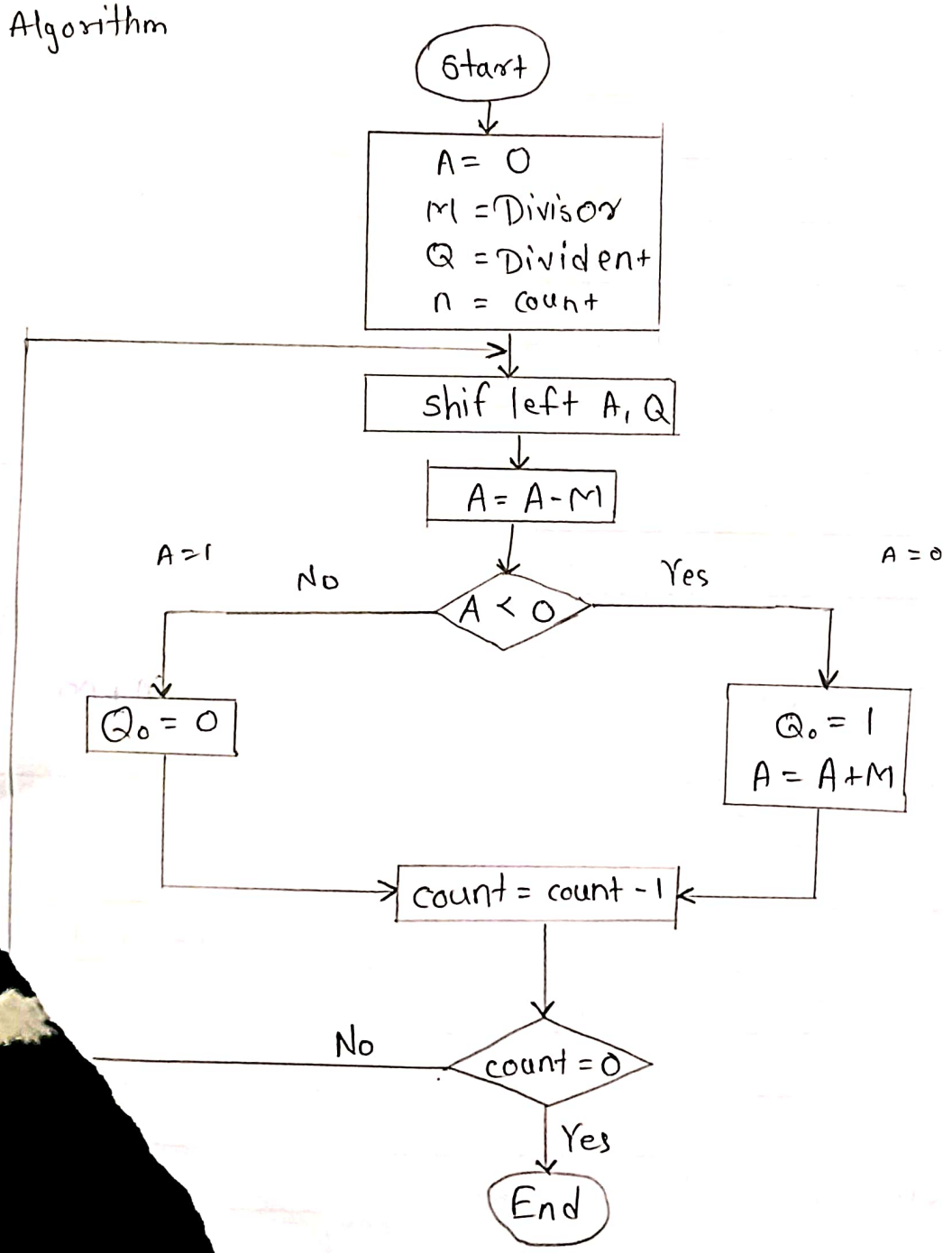
$\therefore (-70)_{10} = (0001000110)_2$

2] Explain the restoring method of binary division algorithm and also give some example.

→

- The restoring method of binary division works by repeatedly shifting and subtracting the divisor from the partial remainder. If the subtraction result is negative, the remainder is restored by adding the divisor back, and the quotient bit is set to 0; otherwise, it is kept and the quotient bit is 1. It is called "restoring" because the remainder is corrected whenever subtraction goes negative.

- Algorithm



Ex., - Divide 7/4 using restore division method.

→ 7 = Q = 0111

4 = 0100 $\xrightarrow{1's}$ 1011

1

M = 1100 ← 2's

A	Q	n	Remark
0000	0111	4	Initialize
0000	111?	3	Shif left AQ
0100	1110		A = A - M
0000	1110		A = A + M
0001	110?	2	Shif left A.Q
0101	1100		A = A - M
0001	1100		A = A + M
0011	100?	1	Shift left A.Q
0111	1000		A = A - M
0011	1000		A = A + M
0111	000?	0	Shif left A.Q
1011	0001		A = A - M
0111	Q		A = A + M
0011			A + M
R			

∴ Quotient = 1

∴ Remainder = 3

2] Explain the [illegible]

3] Show IEEE-754 standard for binary floating point representation for 32-bit single format and 64-bit double format.

→ 1. Single format 32-bit. (single Presision)

31	30 ... 23	22 ... 0
Sign	Exponent	Mentissa

↑

Biased Exponent = exponent value + biased

Biased = 127

e.g., $(34.25)_{10} \longrightarrow 100010.01$

$= 1.0001001 \times 10^5$

Biased exponent = $5+127 = 132 = 10000100$

31	30 ... 23	22 ... 0
S	Exponent	Mentissa
0	10000100	0.0001001

2. 64-bit double format (Double Precision).

Sign	Exponent	Mentissa

↑

Biased Exponent = exponent value + biased

Biased = 1023

eg., $(34.25)_{10} \longrightarrow 100010.01$

$= 1.0001001 \times 10^5$

Biased exponent = $5 + 1023 = 1028 = 10000000100$

S	Exponent	Mentissa
0	10000000100	0.0001001

4] Add $(57)_{10}$ and $(26)_{10}$ in BCD.

→ 57 = 0101 0111

26 = 0010 0110

$$\begin{array}{r} 0101\ 0111 \\ +\ 0010\ 0110 \\ \hline 0111\ 1101 \\ 7\qquad 13 \end{array}$$

$$\begin{array}{r} 0110 \\ \hline 1000\ 0011 \\ 8\qquad 3 \end{array}$$

$\therefore (57)_{10} + (26)_{10} = (83)_{10} = (1000\ 0011)_2$

5] Represent $(52)_{10}$ into excess 3 and Grey Code.

→ $(52)_{10}$ =

1. Excess 3 code

$$\begin{array}{r} 0101\ 0010 \\ 0011\ 0011 \\ \hline 1000\ 0101 \end{array}$$

$\therefore (52)_{10} = 1000\ 0101$

2. Grey code

$$\begin{array}{cccc} 0+1+0+1 & & 0+0+1+0 \\ \downarrow\ \downarrow\ \downarrow\ \downarrow & & \downarrow\ \downarrow\ \downarrow\ \downarrow \\ 0\ 1\ 1\ 1 & & 0\ 0\ 1\ 1 \end{array}$$

$\therefore (52)_{10} = 0111\ 0011$